

Unifying Special Relativity and Quantum Mechanics

Lecture 2 — Completing the Irreps of $SE(2)$; the Poincaré Group, its Structure and Lie Algebra

(Typed version of the handwritten notes, with added explanations in blue boxes)

Added explanation (not in the original notes)

How to read this document. Same conventions as for Lecture 1: normal text and **yellow boxes** are a faithful transcription of the handwritten notes (yellow boxes correspond to the highlighted results in the original); **blue boxes** like this one contain material I added. Page markers [p. N] refer to the handwritten pages, which for this lecture run from 23 to 44.

Where we left off (recap of Lecture 1). For $SE(2)$ we fixed the Casimir eigenvalue p^2 , chose a reference momentum $\vec{p}_{\text{ref}} = (p, 0)^T$, and *defined* the states $|\vec{p}\rangle \equiv \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle$ for $\theta \in [0, 2\pi)$ — this was equation $(\star)$. We showed each $|\vec{p}\rangle$ is a $\vec{P}$ -eigenstate with eigenvalue $\vec{p} = R(\theta)\vec{p}_{\text{ref}}$, and that translations act as a phase. What was still missing: how *rotations* act on a *generic* $|\vec{p}\rangle$ (not just on $|\vec{p}_{\text{ref}}\rangle$), irreducibility, and unitarity. That is where this lecture starts.

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1	Action of rotations on generic states $ \vec{p}\rangle$	[p. 23]	

$$\mathcal{U}(\vec{a}) |\vec{p}\rangle = e^{-i\vec{a}\cdot\vec{p}} |\vec{p}\rangle, \quad \mathcal{U}(\vec{a} + \vec{b}) = \mathcal{U}(\vec{a})\mathcal{U}(\vec{b})$$

The operator is unitary; the group composition rule is verified.

We thus focus on rotation. What is the action of rotations on the states $|\vec{p}\rangle$? $\leftrightarrow$ previously we computed the action of rotations on $|\vec{p}_{\text{ref}}\rangle$; we now consider the case of generic $|\vec{p}\rangle$.

The result is pretty obvious, let us look at it. We consider a rotation $\mathcal{U}(\phi)$ acting on $|\vec{p}\rangle$. First, we act on $\mathcal{U}(\phi)|\vec{p}\rangle$ with $\vec{P}$:

$$\begin{aligned}\vec{P}\mathcal{U}(\phi)|\vec{p}\rangle &= \mathcal{U}(\phi) \underbrace{\mathcal{U}(\phi)^{-1}\vec{P}\mathcal{U}(\phi)}_{R(\phi)\vec{P}}|\vec{p}\rangle = \mathcal{U}(\phi)R(\phi)\vec{P}|\vec{p}\rangle \\ &= \mathcal{U}(\phi)R(\phi)\vec{p}|\vec{p}\rangle = \mathcal{U}(\phi)\vec{p}'|\vec{p}\rangle = \vec{p}'\mathcal{U}(\phi)|\vec{p}\rangle,\end{aligned}$$

defining $\vec{p}' \equiv R(\phi)\vec{p}$.

$\rightsquigarrow \mathcal{U}(\phi)|\vec{p}\rangle$ is eigenstate of $\vec{P}$ with eigenvalue $\vec{p}' = R(\phi)\vec{p} = R(\phi)R(\theta)\vec{p}_{\text{ref}} = R(\phi + \theta)\vec{p}_{\text{ref}}$.

Added explanation (not in the original notes)

The key identity used here, $\mathcal{U}(\phi)^{-1}P^k\mathcal{U}(\phi) = (R(\phi))_{kj}P^j$, is exactly the one derived at the end of Lecture 1 (p. 21), valid in *all* representations: the operators P^k transform as the components of a vector under conjugation by a rotation. Also note the step from $R(\phi)R(\theta)$ to $R(\phi + \theta)$: it uses the fact that $SO(2)$ is *abelian* — rotations in the plane about the same point simply add their angles. This is special to two dimensions and will *not* carry over to the Lorentz group later in this lecture.

From our definition ($\star$), we actually know already that also the state [p. 24]

$$|\vec{p}'\rangle = \mathcal{U}(\theta + \phi)|\vec{p}_{\text{ref}}\rangle$$

is, by definition, eigenstate of $\vec{P}$ with eigenvalue $\vec{p}'$!

$\rightsquigarrow \mathcal{U}(\phi)|\vec{p}\rangle$ and $|\vec{p}'\rangle = \mathcal{U}(\theta + \phi)|\vec{p}_{\text{ref}}\rangle$ are, if not equal, at least proportional to each other.

We find the proportionality as follows:

$$\mathcal{U}(\phi)|\vec{p}\rangle = \mathcal{U}(\phi)\mathcal{U}(\theta)|\vec{p}_{\text{ref}}\rangle = \mathcal{U}(\phi + \theta) \underbrace{[\mathcal{U}(\phi + \theta)^{-1}\mathcal{U}(\phi)\mathcal{U}(\theta)]}_{\text{this is the identity operator}}|\vec{p}_{\text{ref}}\rangle \stackrel{\circledast}{=}$$

This is the identity operator: the first rotation brings $\vec{p}_{\text{ref}}$ into $R(\theta)\vec{p}_{\text{ref}}$, the second brings $R(\theta)\vec{p}_{\text{ref}}$ into $R(\phi)R(\theta)\vec{p}_{\text{ref}}$, but the third one brings it back to $\vec{p}_{\text{ref}}$.

$$\stackrel{\circledast}{=} \mathcal{U}(\phi + \theta)|\vec{p}_{\text{ref}}\rangle = |\vec{p}'\rangle.$$

From the comparison, we thus find

$$|\vec{p}'\rangle = \mathcal{U}(\phi)|\vec{p}\rangle, \quad \mathcal{U}(\phi + \theta) = \mathcal{U}(\phi)\mathcal{U}(\theta), \quad \text{with } \vec{p}' = R(\phi)\vec{p}.$$

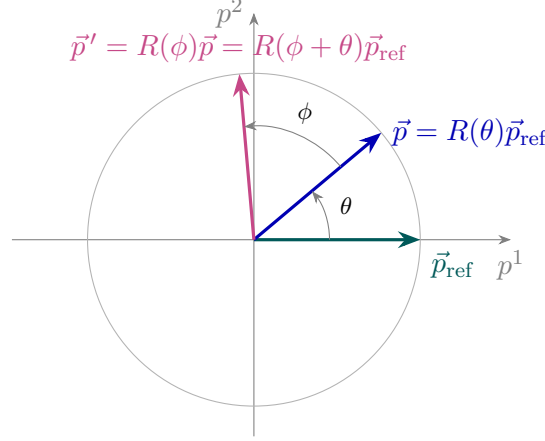
The group composition rule is verified.

Added explanation (not in the original notes)

Why “if not equal, at least proportional”? Both $\mathcal{U}(\phi)|\vec{p}\rangle$ and $|\vec{p}'\rangle$ are eigenstates of $\vec{P}$ with the *same* eigenvalue $\vec{p}'$. In this representation the spectrum of $\vec{P}$ on the circle is non-degenerate (one state per momentum value), so two eigenstates with the same eigenvalue can differ at most by a multiplicative constant — a phase, since both are normalized in the δ -function sense. The explicit computation then shows that this phase is exactly 1: no extra phase factor appears. (In more general induced representations — e.g. for the Poincaré group with spin — the analogous “sandwich” $\mathcal{U}^{-1}\mathcal{U}\mathcal{U}$ is an element of the *little group* and acts non-trivially on internal indices; here the little group of $\vec{p}_{\text{ref}}$ is trivial, which is why the states

carry no extra label besides $\vec{p}$.)

Note also the group-theoretic content of the identity-operator argument: the operator $\mathcal{U}(\phi + \theta)^{-1} \mathcal{U}(\phi) \mathcal{U}(\theta)$ represents the group element $R(\phi + \theta)^{-1} R(\phi) R(\theta)$, which is the identity element of $SO(2)$; since a representation is a homomorphism, it must be represented by $\mathbb{1}_V$.



(Figure added for clarity — not in the original notes.)

2 The representation: irreducibility

[p. 25]

■ We thus have a *representation* in the sense that, acting on the states $|\vec{p}\rangle$, the group transformations $\mathcal{U}(\vec{a})$ and $\mathcal{U}(\theta)$ verify the group transformation laws.

↔ The representation is irreducible only once we include all states

$$\left\{ |\vec{p}\rangle = \mathcal{U}(\theta) |\vec{p}_{\text{ref}}\rangle, \quad \theta \in [0, 2\pi) \right\}_{p=|\vec{p}| \text{ fixed}}$$

since they can all be obtained with the action of an appropriate rotation.

↔ On the other hand, it is not possible to change the value of $|\vec{p}|$ acting with group transformations, and every irrep will be labeled by the corresponding value of p fixed.

Added explanation (not in the original notes)

Unpacking the two statements: (i) starting from any state of the circle you can reach any other by a rotation, so no proper subspace spanned by a subset of the $|\vec{p}\rangle$'s is invariant — that is irreducibility; (ii) translations multiply by a phase and rotations move along the circle, but nothing changes the radius p — consistent with $|\vec{P}|^2$ being a *Casimir*: it commutes with everything, so it is constant ($= p^2$) throughout the irrep and serves as its label. Distinct $p > 0$ give inequivalent irreps. (For $p = 0$ the circle degenerates to a point, translations act trivially, and the irreps are the one-dimensional representations of $SO(2)$ labeled by an integer — not discussed in the notes.)

↪ It is possible to show that the group transformation law remains verified even if we consider the combination of two generic group elements $\mathcal{U}(\theta, \vec{a})$ and $\mathcal{U}(\phi, \vec{b})$:

$$\mathcal{U}(\theta, \vec{a}) \mathcal{U}(\phi, \vec{b}) = \mathcal{U}(\theta + \phi, R(\theta)\vec{b} + \vec{a}),$$

if we consider

$$\mathcal{U}(\theta, \vec{a}) = \mathcal{U}(\vec{a}) \mathcal{U}(\theta) = e^{-i\vec{a} \cdot \vec{P}} e^{-i\theta J}.$$

Added explanation (not in the original notes)

This is exactly the $SE(2)$ composition rule of Lecture 1 (p. 6), $\mathcal{E}(R_2, \vec{a}_2)\mathcal{E}(R_1, \vec{a}_1) = \mathcal{E}(R_2 R_1, R_2 \vec{a}_1 + \vec{a}_2)$, now verified at the level of the *operators on the Hilbert space*. Sketch of the check: move $\mathcal{U}(\phi, \vec{b}) = \mathcal{U}(\vec{b})\mathcal{U}(\phi)$ next to $\mathcal{U}(\vec{a})\mathcal{U}(\theta)$ and use the conjugation identity to push $\mathcal{U}(\theta)$ through $\mathcal{U}(\vec{b})$: $\mathcal{U}(\theta)\mathcal{U}(\vec{b}) = \mathcal{U}(R(\theta)\vec{b})\mathcal{U}(\theta)$. Collecting translations and rotations separately gives $\mathcal{U}(R(\theta)\vec{b} + \vec{a})\mathcal{U}(\theta + \phi)$, which is the claim.

3 The representation is unitary

[p. 26]

First, we equip the vector space with a scalar product

$$\langle \vec{p} | \vec{q} \rangle = \delta(\vec{p} - \vec{q})$$

and a completeness relation

$$\int d^2 p \, |\vec{p}\rangle \langle \vec{p}| = \mathbb{K}.$$

We thus have

$$|\vec{p}'\rangle = \mathcal{U}(\theta) |\vec{p}\rangle, \quad \langle \vec{p}'| = \langle \vec{p}| \mathcal{U}(\theta)^\dagger,$$

and

$$\mathcal{U}(\theta)\mathcal{U}(\theta)^\dagger = \int d^2 p \, \mathcal{U}(\theta) |\vec{p}\rangle \langle \vec{p}| \mathcal{U}(\theta)^\dagger = \int d^2 p \, |\vec{p}'\rangle \langle \vec{p}'| \stackrel{\textcircled{*}}{=} \mathbb{K}$$

with $\vec{p}' = R(\theta)\vec{p}$. Since rotations have $\det R(\theta) = +1$, it follows that

$$d^2 p' = d^2 p$$

and

$$\stackrel{\textcircled{*}}{=} \int d^2 p' \, |\vec{p}'\rangle \langle \vec{p}'| = \mathbb{K} \quad \rightsquigarrow \quad \mathcal{U}(\theta)\mathcal{U}(\theta)^\dagger = \mathbb{K}.$$

Added explanation (not in the original notes)

(The handwritten page writes $\int d^3 p$; for $SE(2)$ momentum space is two-dimensional, so the measure is $d^2 p$ — the same $\mathbb{R}^3/\mathbb{R}^2$ slip already flagged in Lecture 1, p. 18. Transcribed here as $d^2 p$.)

Logic of the unitarity proof. The change of variables $\vec{p} \rightarrow \vec{p}' = R(\theta)\vec{p}$ has Jacobian $|\det R(\theta)| = 1$, so the integration measure is invariant; then the completeness relation re-assembles itself and gives $\mathcal{U}\mathcal{U}^\dagger = \mathbb{K}$. Together with the analogous statement for translations (whose action is a pure phase, manifestly preserving norms), this shows every group element is represented by a unitary operator — which is what “unitary representation” means. Physically: probabilities $|\langle \phi | \psi \rangle|^2$ are preserved under rotations and translations of the system.

Interpretation

We arrive at the same equations from where we started, but we gain a new perspective:

$$\hookrightarrow \text{meaning} \quad \vec{P} |\vec{p}\rangle = \vec{p} |\vec{p}\rangle, \quad \left| \vec{P} \right|^2 |\vec{p}\rangle = p^2 |\vec{p}\rangle, \quad \langle \vec{p} | \vec{q} \rangle = \delta(\vec{p} - \vec{q}), \quad \int d^2 p \, |\vec{p}\rangle \langle \vec{p}| = \mathbb{K}.$$

[p. 27]

The description of a free particle with fixed energy in NRQM follows the unitary irreps of the group $SE(2)$.

↓

Not a surprise: the system is invariant under spatial translations (conservation of momentum) and rotations (conservation of the angular momentum perpendicular to the plane).

Prompted by this finding, we take the following route:

To describe a relativistic free particle, we look for unitary, irreducible representations of the symmetry group of special relativity: the Poincaré group.

Added explanation (not in the original notes)

This is the pivot point of the whole course — Wigner’s programme (1939): *particles = unitary irreducible representations of the spacetime symmetry group*. Everything done so far for $SE(2)$ was a warm-up in which each ingredient has a relativistic counterpart:

$$\begin{array}{ll}
 SE(2) = T(2) \rtimes SO(2) & \longleftrightarrow \text{Poincaré} = T(1,3) \rtimes \text{Lorentz} \\
 \text{Casimir } |\vec{P}|^2 = p^2 & \longleftrightarrow \text{Casimir } P^\mu P_\mu = m^2 \text{ (and spin)} \\
 \text{circle } |\vec{p}| = p & \longleftrightarrow \text{mass shell } E^2 - |\vec{p}|^2 = m^2 \\
 \text{reference vector } \vec{p}_{\text{ref}} & \longleftrightarrow \text{rest-frame momentum } (m, \vec{0}) \\
 \text{rotations moving } \vec{p}_{\text{ref}} & \longleftrightarrow \text{boosts/rotations moving } (m, \vec{0})
 \end{array}$$

The rest of this lecture builds the group-theory toolbox for the Poincaré group, following the same sequence used for $E(m)$ in Lecture 1: definition $\rightarrow$ parameter counting $\rightarrow$ composition/inverse/factorization $\rightarrow$ connected components $\rightarrow$ Lie algebra $\rightarrow$ generators and commutators.

4 The Poincaré group: structure and properties

[p. 28]

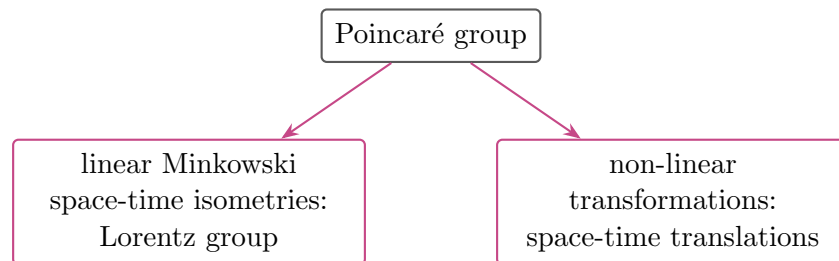
Ten-dimensional group of Minkowski space-time isometries.

$\hookrightarrow$ The space-time interval between two events is left invariant. Given $(t, \vec{x})$ and $(t_0, \vec{x}_0)$,

$$\Delta s^2 \equiv (t - t_0)^2 - |\vec{x} - \vec{x}_0|^2 \text{ is invariant.}$$

$\hookrightarrow$ Same as the Euclidean group, but the underlying metric is *Lorentzian* and not Euclidean.

We exploit our previous understanding of the Euclidean group.



Added explanation (not in the original notes)

The dimension counting 10 anticipates what follows: 4 parameters for spacetime translations + 6 for the Lorentz group (computed on p. 30). Compare with the Euclidean formula of Lecture 1, $\dim E(m) = m + \frac{m(m-1)}{2}$: formally with $m = 4$ this gives $4 + 6 = 10$ — the counting is the same, only the notion of “distance” preserved is different (Lorentzian instead of Euclidean).

[p. 29] ■ Non-linear translations are transformations of the form

$$x^\mu \rightarrow x'^\mu = x^\mu + a^\mu$$

identified by 4 real parameters a^μ .

■ Linear Lorentz transformations leave invariant the quadratic form $t^2 - |\vec{x}|^2$.

If we introduce the four-component vector $\begin{pmatrix} t \\ \vec{x} \end{pmatrix}$ and the metric matrix

$$g \equiv \begin{pmatrix} 1 & 0 \\ 0 & -\mathbb{K} \end{pmatrix},$$

we have

$$(t, \vec{x}) g \begin{pmatrix} t \\ \vec{x} \end{pmatrix} = (t, \vec{x}) \begin{pmatrix} t \\ -\vec{x} \end{pmatrix} = t^2 - |\vec{x}|^2.$$

Consequently, a linear transformation of the form

$$\begin{pmatrix} t \\ \vec{x} \end{pmatrix} \rightarrow \begin{pmatrix} t' \\ \vec{x}' \end{pmatrix} \equiv \Lambda \begin{pmatrix} t \\ \vec{x} \end{pmatrix}$$

leaves $t^2 - |\vec{x}|^2$ invariant if

$$(t, \vec{x}) g \begin{pmatrix} t \\ \vec{x} \end{pmatrix} \rightarrow (t, \vec{x}) \Lambda^T g \Lambda \begin{pmatrix} t \\ \vec{x} \end{pmatrix} \stackrel{!}{=} (t, \vec{x}) g \begin{pmatrix} t \\ \vec{x} \end{pmatrix} \rightsquigarrow \boxed{\Lambda^T g \Lambda = g}$$

The Lorentz group can be indeed be defined by [p. 30]

$$O(1, 3) = \left\{ \Lambda \in GL(4, \mathbb{R}) \mid \Lambda^T g \Lambda = g \right\} \quad \text{Lorentz group, 6 real parameters.}$$

Why 6? $\Lambda \in GL(4, \mathbb{R}) \rightsquigarrow 16$ real entries. However, not all are independent, since subject to the constraint $\Lambda^T g \Lambda = g$. How many constraining equations? In principle, every constraining equation eliminates one parameter. In principle, $\Lambda^T g \Lambda = g$ is a matrix set of $4 \times 4 = 16$ equations. Not all of them, however, are independent. In fact, if we take the transpose:

$$(\Lambda^T g \Lambda)^T = \Lambda^T g^T \Lambda = \Lambda^T g \Lambda = g$$

$\rightsquigarrow$ it is a *symmetric* set of equations, and only $\frac{4(4+1)}{2} = 10$ are independent.

$$\underbrace{16}_{\text{initial DOF}} - \underbrace{10}_{\text{number of independent constraints}} = 6 \text{ real parameters.}$$

Added explanation (not in the original notes)

This is the same argument used in Lecture 1 (p. 5) for $O(m)$: there $R^T R = \mathbb{K}$ gave $m^2 - \frac{m(m+1)}{2} = \frac{m(m-1)}{2}$ parameters. Here g replaces $\mathbb{K}$ but the constraint matrix $\Lambda^T g \Lambda$ is still symmetric, so the counting is identical with $m = 4$: $16 - 10 = 6$. The 6 parameters will turn out to be 3 rotation angles + 3 boost rapidities.

5 Tensor notation

[p. 31]

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & -\mathbb{K} \end{pmatrix}, \quad x^\mu = (x^0 \equiv t, \vec{x}), \quad x_\mu = g_{\mu\nu} x^\nu = (x^0 \equiv t, -\vec{x}), \quad \delta^\mu_\nu = \begin{pmatrix} 1 & 0 \\ 0 & \mathbb{K} \end{pmatrix}.$$

The inverse metric is defined by

$$(g^{-1})^{\mu\nu} g_{\nu\rho} = \delta_{\rho}^{\mu} \quad \rightsquigarrow \quad (g^{-1})^{\mu\nu} = \begin{pmatrix} 1 & 0 \\ 0 & -\mathbb{I} \end{pmatrix}.$$

A Lorentz transformation is described by

$$x^{\mu} \rightarrow x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu}$$

such that

$$g_{\mu\nu} x^{\mu} x^{\nu} = (t^2 - |\vec{x}|^2) \rightarrow g_{\mu\nu} x'^{\mu} x'^{\nu} = g_{\mu\nu} \Lambda^{\mu}_{\rho} \Lambda^{\nu}_{\sigma} x^{\rho} x^{\sigma} \stackrel{!}{=} g_{\rho\sigma} x^{\rho} x^{\sigma}$$

$$g_{\mu\nu} \Lambda^{\mu}_{\rho} \Lambda^{\nu}_{\sigma} = g_{\rho\sigma} \quad \text{analog of } \Lambda^T g \Lambda = g \text{ but in tensor notation.}$$

$\triangle$ The transpose is defined by the relation

$$(\Lambda^T)_{\mu}^{\nu} = \Lambda^{\nu}_{\mu}$$

while for the inverse Λ^{-1} we write

$$\Lambda^T g \Lambda = g \rightsquigarrow g^{-1} \Lambda^T g = \Lambda^{-1} \rightsquigarrow (\Lambda^{-1})^{\mu}_{\nu} = (g^{-1} \Lambda^T g)^{\mu}_{\nu} = (g^{-1})^{\mu\rho} (\Lambda^T)_{\rho}^{\sigma} g_{\sigma\nu} =$$

[p. 32]

$$= (g^{-1})^{\mu\rho} \Lambda^{\sigma}_{\rho} g_{\sigma\nu} = \Lambda^{\mu}_{\nu}.$$

$$(\Lambda^{-1})^{\mu}_{\nu} = \Lambda^{\mu}_{\nu}$$

Added explanation (not in the original notes)

Reading the index gymnastics. Λ_{ν}^{μ} means: take Λ^{μ}_{ν} and lower the first index / raise the second with the metric, $\Lambda_{\nu}^{\mu} = g_{\nu\alpha} \Lambda^{\alpha}_{\beta} (g^{-1})^{\beta\mu}$. The boxed result is the Lorentzian analogue of $R^{-1} = R^T$ for orthogonal matrices: the inverse is the “metric transpose”. Because the metric has minus signs, this is *not* the naive matrix transpose: concretely, space-space components transpose as usual, while time-space components pick up an extra sign (e.g. for a boost, $(\Lambda^{-1})^0_i = -\Lambda^0_i$, consistent with inverting a boost by flipping its velocity). The position of the indices (which one is up, which one is down, and their left-right order) is doing all the bookkeeping.

Acting on four-vectors, a generic Poincaré transformation reads

$$x^{\mu} \rightarrow x'^{\mu} = \Lambda^{\mu}_{\nu} x^{\nu} + a^{\mu} \quad \text{this is the “defining representation” of Poincaré.}$$

We write a generic Poincaré transformation in the form

$$P(\Lambda, a)$$

where Λ is an element of the Lorentz group and a^{μ} is a four-vector identifying a space-time translation. The composition rule reads

$$P(\Lambda_2, a_2) P(\Lambda_1, a_1) = P(\Lambda_2 \Lambda_1, \Lambda_2 a_1 + a_2)$$

(we can find it by looking at the defining representation). [p. 33]

The neutral element is

$$e = P(\mathbb{K}, 0)$$

where $\mathbb{K}$ is the neutral element of the Lorentz group; it is $\mathbb{K}_{4 \times 4}$ in the defining representation. The inverse element is

$$P(\Lambda, a)^{-1} = P(\Lambda^{-1}, -\Lambda^{-1}a).$$

We have the factorization property

$$P(\Lambda, a) = P(\mathbb{K}, a) P(\Lambda, 0) \quad \triangle \text{ the order is important!}$$

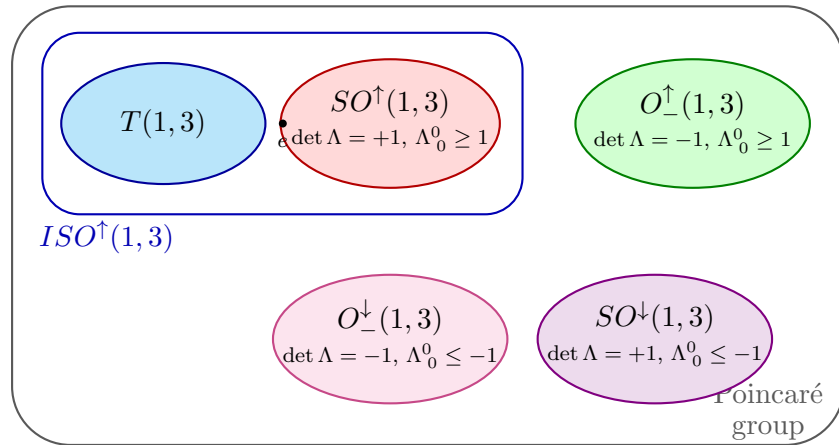
It follows from the composition rule.

$$P(\Lambda, a) = \underbrace{P(\mathbb{K}, a)}_{\substack{T(1,3) \equiv T(a), \text{ subgroup of} \\ \text{space-time translations} \\ T(a)T(b)=T(a+b)}} \underbrace{P(\Lambda, 0)}_{\substack{O(1,3) \equiv L(\Lambda), \text{ subgroup of} \\ \text{Lorentz transformations}}}$$

Added explanation (not in the original notes)

Every formula in this section is the word-for-word analogue of the $E(m)$ formulas of Lecture 1 with the dictionary $R \rightarrow \Lambda$, $\vec{a} \rightarrow a^\mu$, $\mathcal{E} \rightarrow P$: composition (L1 p. 6), inverse (p. 7), factorization and its order-sensitivity (p. 8), and the normality of the translation subgroup (p. 7, same computation verbatim). So the Poincaré group is also a semi-direct product, $\text{Poincaré} = T(1,3) \rtimes O(1,3)$, and one could equally embed it in $GL(5, \mathbb{R})$ with homogeneous coordinates. The notes do not repeat these derivations — they were the whole point of doing $E(m)$ first.

6 Connected components; the identity component $ISO^\uparrow(1,3)$ [p. 34]



$ISO^\uparrow(1,3)$: identity component of the Poincaré group (transformations continuously connected to the identity).

Added explanation (not in the original notes)

Where the four Lorentz pieces come from. Two discrete, continuous-path-proof labels classify Λ : (i) $\det \Lambda = \pm 1$ (from $(\det \Lambda)^2 = 1$, exactly as for $O(m)$); (ii) the sign of Λ_0^0 . Indeed the 00-component of $g_{\mu\nu}\Lambda^\mu_\rho\Lambda^\nu_\sigma = g_{\rho\sigma}$ gives $(\Lambda_0^0)^2 = 1 + \sum_i (\Lambda_i^0)^2 \geq 1$, so $\Lambda_0^0 \geq 1$ or ≤ -1 , with no way to pass continuously from one branch to the other. Transformations

with $\Lambda_0^0 \geq 1$ are called *orthochronous* ($\uparrow$): they preserve the direction of time. The four sign combinations give the four components drawn above; representatives of the three non-identity ones are parity $P = \text{diag}(1, -1, -1, -1) \in O_-^\uparrow$, time reversal $T = \text{diag}(-1, 1, 1, 1) \in O_-^\downarrow$, and $PT = -\mathbb{K} \in SO^\downarrow$. Only $SO^\uparrow(1, 3)$ (proper, orthochronous) is a connected subgroup containing e — the Lorentzian analogue of the $SO(m)$ -vs- $O_-(m)$ split of Lecture 1 (p. 10), except that the Lorentzian signature doubles the number of pieces from 2 to 4. As in Lecture 1, only the identity component is captured by exponentiating the Lie algebra, which is what follows.

We now have to consider the Lie algebra of the Poincaré group. Because of the factorization property, we focus separately on Lorentz transformations and space-time translations.

7 The Lie algebra of the Lorentz group

We consider the defining representation. We look for a real 4×4 matrix X such that [p. 35]

$$e^{tX} \in SO^\uparrow(1, 3) \quad \forall t \in \mathbb{R}.$$

This condition means that

$$(e^{tX})^T g (e^{tX}) \stackrel{!}{=} g \rightsquigarrow \underbrace{g^{-1} (e^{tX})^T g}_{g^{-1} e^{tX^T} g = e^{t g^{-1} X^T g}} = (e^{tX})^{-1} = e^{-tX},$$

that is

$$e^{t g^{-1} X^T g} = e^{-tX} \rightsquigarrow \boxed{X^T g = -gX}$$

We thus get

$$\mathfrak{so}(1, 3) = \left\{ X \in M(4, \mathbb{R}) \mid X^T g = -gX \right\} \quad \text{Lie algebra of the Lorentz group.}$$

Added explanation (not in the original notes)

Two small steps used silently: $g^{-1} e^{tX} g = e^{g^{-1} tX g}$ (conjugation commutes with the power series) and equality of the exponentials for all t forces equality of the exponents (differentiate at $t = 0$). The condition $X^T g = -gX$ says gX is antisymmetric, i.e. X is “antisymmetric with respect to the metric g ” — the Lorentzian deformation of the Euclidean condition $Y^T = -Y$ found in Lecture 1 (p. 12). Setting $g \rightarrow \mathbb{K}$ recovers exactly that case. Note this also forces $\text{tr } X = 0$, hence $\det e^{tX} = +1$: the exponential automatically lands in the proper part of the group, as anticipated.

We now look for a basis in the algebra to identify the generators. Consider a generic matrix

$$X = \left(\begin{array}{c|c} A & \vec{b}^T \\ \hline \vec{c} & D \end{array} \right) \quad (A: \text{number}, D: 3 \times 3 \text{ block})$$

and write the condition $X^T g = -gX$ as [p. 36]

$$\begin{aligned} \left(\begin{array}{cc} A & \vec{c}^T \\ \vec{b} & D^T \end{array} \right) \left(\begin{array}{cc} 1 & 0 \\ 0 & -\mathbb{K} \end{array} \right) &\stackrel{!}{=} - \left(\begin{array}{cc} 1 & 0 \\ 0 & -\mathbb{K} \end{array} \right) \left(\begin{array}{cc} A & \vec{b}^T \\ \vec{c} & D \end{array} \right) \\ \left(\begin{array}{cc} A & -\vec{c}^T \\ \vec{b} & -D^T \end{array} \right) &= - \left(\begin{array}{cc} A & \vec{b}^T \\ -\vec{c} & -D \end{array} \right) \rightsquigarrow \begin{cases} A = -A \Rightarrow A = 0 \\ \vec{c} = \vec{b} \\ D^T = -D \end{cases} \end{aligned}$$

Consequently, X belongs to the algebra only if it has the form

$$X = \begin{pmatrix} 0 & \vec{b}^T \\ \vec{b} & D \end{pmatrix}, \quad \text{with } D = -D^T.$$

$$\mathfrak{so}(1,3) = \left\{ X = \begin{pmatrix} 0 & \vec{b}^T \\ \vec{b} & D \end{pmatrix} \mid \vec{b} \in \mathbb{R}^3 \wedge D = -D^T \right\}$$

It is now easy to extract the generators. The generic form of X is the following:

$$X = \left(\begin{array}{c|ccc} 0 & b^1 & b^2 & b^3 \\ \hline b^1 & 0 & -a & c \\ b^2 & a & 0 & -d \\ b^3 & -c & d & 0 \end{array} \right) \quad \begin{array}{l} b^1, b^2, b^3 \in \mathbb{R} \\ a, c, d \in \mathbb{R} \end{array}$$

$\rightsquigarrow$ 6 basis elements: three generators are needed to generate the 3×3 block, and three generators are needed to generate the first row/column.

Added explanation (not in the original notes)

Structure of this matrix at a glance: the antisymmetric 3×3 block D is exactly an element of $\mathfrak{so}(3)$ (compare $Y^T = -Y$ for $\mathfrak{so}(2)$ in Lecture 1, p. 13) — it will generate *rotations*; the symmetric first row/column $\vec{b}$ has no Euclidean analogue — it is there because of the relative minus sign in g , and it will generate *boosts*, which mix time with space. (Small notational slip in the original: the parameters of D are introduced as “ a, b, c ” on p. 36 but the matrix uses a, c, d , and p. 37 then writes the linear combination with coefficients b, c, a matching T^1, T^2, T^3 . Transcribed here as a, c, d for the matrix, consistently with the entries actually drawn.)

[p. 37]

$$\begin{aligned} T^1 &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, & T^2 &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, & T^3 &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \\ T^4 &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, & T^5 &= \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, & T^6 &= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

With these definitions, we have

$$X = b^1 T^4 + b^2 T^5 + b^3 T^6 + c T^2 + a T^3 + b T^1 \quad (\text{generic } X \text{ written as linear combination of generators}).$$

We switch to physicist's convention:

$$X = \alpha_A T^A = -i\alpha_A (iT^A) \quad (\text{“physicist's convention”}).$$

$$J_{4\text{vec}}^1 \equiv iT^1 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \end{pmatrix}, \quad J_{4\text{vec}}^2 \equiv iT^2 = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & -i & 0 & 0 \end{pmatrix}, \quad J_{4\text{vec}}^3 \equiv iT^3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

[p. 38] By direct computation, we get

$$[J^i, J^j] = i \varepsilon_{ijk} J^k$$

Lie algebra $\mathfrak{so}(3) \cong \mathfrak{su}(2)$: commutation relations of the angular momentum.

The other three generators are

$$K_{4\text{vec}}^1 \equiv iT^4 = \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_{4\text{vec}}^2 \equiv iT^5 = \begin{pmatrix} 0 & 0 & i & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad K_{4\text{vec}}^3 \equiv iT^6 = \begin{pmatrix} 0 & 0 & 0 & i \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix}.$$

By direct computation:

$$[K^i, K^j] = -i \varepsilon_{ijk} J^k, \quad [J^i, K^j] = i \varepsilon_{ijk} K^k.$$

Obs.: we worked out the explicit form of the generators in the representation acting on 4-vectors (that's why we put a subscript "4vec"). [p. 39] However, the Lie brackets among them are valid in *all* representations, so we wrote them without a specific subscript.

Added explanation (not in the original notes)

Physical reading of the three brackets. $[J^i, J^j] = i \varepsilon_{ijk} J^k$: rotations close among themselves — the rotation subgroup $SO(3)$ sits inside the Lorentz group. $[J^i, K^j] = i \varepsilon_{ijk} K^k$: $\vec{K}$ transforms as a *vector* under rotations (same pattern as $[J, P^i]$ for $SE(2)$ in Lecture 1, p. 14). $[K^i, K^j] = -i \varepsilon_{ijk} J^k$: two boosts do *not* compose into a boost — their commutator is a rotation. This single minus-sign-laden bracket is responsible for Thomas precession and, as noted on p. 42 below, for the fact that boosts alone do not form a subgroup. It is also the only bracket that distinguishes $\mathfrak{so}(1, 3)$ from the Euclidean algebra $\mathfrak{so}(4)$ (where the analogous bracket has $+i$).

We now use the exponential map to relate the algebra to the group elements. We write

$$D(\vec{\theta}, \vec{\eta}) = \exp(-i \vec{\theta} \cdot \vec{J} - i \vec{\eta} \cdot \vec{K})$$

6 parameters $\vec{\theta}$ and $\vec{\eta}$ are needed to identify a Lorentz transformation; this is the exponential of a real linear combination of generators, and it is valid in *all representations*. If we consider the explicit form of the generators in the four-vector representation, we get the “usual” Lorentz transformations acting on four-vectors. We write

$$\Lambda(\vec{\theta}, \vec{\eta}) = \exp(-i \vec{\theta} \cdot \vec{J}_{4\text{vec}} - i \vec{\eta} \cdot \vec{K}_{4\text{vec}}).$$

8 Rotations

[p. 40]

Transformations generated by $\vec{J}$. We write them as

$$D(\vec{\theta}) \equiv D(\vec{\theta}, \vec{\eta} = 0) = \exp(-i \vec{\theta} \cdot \vec{J}) = \exp(-i \theta \hat{n} \cdot \vec{J}) \equiv D(\theta, \hat{n}).$$

■ Rotations form a subgroup of the Lorentz group: the rotation group $SO(3)$. It is also possible to organize the three parameters identifying a rotation into a rotation angle θ and an axis $\hat{n}$ around which we rotate.

In the defining representation, we use the notation

$$\Lambda_R(\vec{\theta}) = \exp(-i\vec{\theta} \cdot \vec{J}_{4\text{vec}}) = \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & R(\vec{\theta}) \end{array} \right) = \exp(-i\theta \hat{n} \cdot \vec{J}_{4\text{vec}}) \equiv \Lambda_R(\theta, \hat{n}).$$

Example #1. We consider a rotation around the $\hat{z}$ -axis by some angle θ . We write [p. 41]

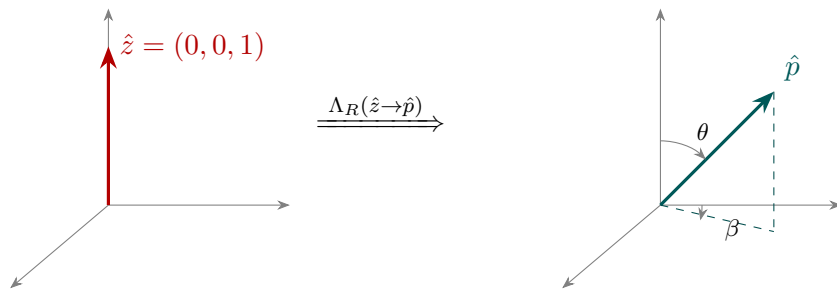
$$\Lambda_R(\theta \hat{z}) = \exp(-i\theta J_{4\text{vec}}^3) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta & 0 \\ 0 & \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad \text{computed via } \sum_{k=0}^{\infty} \frac{(-i\theta)^k}{k!} (J_{4\text{vec}}^3)^k.$$

Added explanation (not in the original notes)

(As in Lecture 1, the handwritten matrix shows both off-diagonal entries as $\sin \theta$; the standard counter-clockwise rotation has $-\sin \theta$ in the $(1, 2)$ -block position, as written above.) How the series resums: with $M \equiv -iJ_{4\text{vec}}^3$, the nonzero 2×2 block of M satisfies $M^2 = -\mathbb{1}$, so even powers give $(-1)^k \mathbb{1}$ and odd powers $(-1)^k M$; the series splits into $\cos \theta \mathbb{1} + \sin \theta M$ acting in the xy -block, while the t and z directions are untouched. The block form $\text{diag}(1, R(\vec{\theta}))$ makes explicit that a rotation leaves time alone — which is why rotations survive unchanged from the Euclidean world into the Lorentz group.

Example #2. Consider the transformation that brings the unit vector $\hat{z} = (0, 0, 1)$ into $\hat{p}$. This is a rotation that we indicate with $\Lambda_R(\hat{z} \rightarrow \hat{p}) \hookrightarrow$ its explicit expression will depend on θ and β (no need to write it explicitly). This rotation acts as follows:

$$\Lambda_R(\hat{z} \rightarrow \hat{p}) \begin{pmatrix} E \\ 0 \\ 0 \\ |\vec{p}| \end{pmatrix} = \begin{pmatrix} E \\ |\vec{p}| \cos \beta \sin \theta \\ |\vec{p}| \sin \beta \sin \theta \\ |\vec{p}| \cos \theta \end{pmatrix}.$$



Added explanation (not in the original notes)

Here θ is the polar angle of $\hat{p}$ measured from the z -axis and β its azimuthal angle in the xy -plane, so the spatial part of the result is just $|\vec{p}| \hat{p}$ written in spherical components — the rotation takes the momentum $|\vec{p}| \hat{z}$ and points it along $\hat{p}$, leaving the energy untouched. This innocent-looking example is a key building block for the Poincaré irreps: the *standard* way to reach a general momentum from the reference one will be “boost along $\hat{z}$, then rotate $\hat{z} \rightarrow \hat{p}$ ” — the relativistic version of “rotate $\vec{p}_{\text{ref}}$ ” from the $SE(2)$ construction.

9 Boosts

[p. 42]

Transformations generated by $\vec{K}$. We write them as

$$D(\vec{\eta}) \equiv D(\vec{0}, \vec{\eta}) = \exp(-i \vec{\eta} \cdot \vec{K}) = \exp(-i \eta \hat{p} \cdot \vec{K}) \equiv D(\eta, \hat{p}),$$

where η is called “*rapidity*” and $\hat{p}$ is the axis along which we boost.

■ Boosts do not form a subgroup (algebra not closed).

In the defining representation, we write

$$\Lambda_B(\eta, \hat{p}) = \exp(-i \eta \hat{p} \cdot \vec{K}_{4\text{vec}}).$$

Example. We boost along $\hat{z}$ with some rapidity η :

$$\Lambda_B(\eta, \hat{z}) = \exp(-i \eta K_{4\text{vec}}^3) = \begin{pmatrix} \cosh \eta & 0 & 0 & \sinh \eta \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \sinh \eta & 0 & 0 & \cosh \eta \end{pmatrix} \quad \text{computed via } \sum_{k=0}^{\infty} \frac{(-i \eta)^k}{k!} (K_{4\text{vec}}^3)^k.$$

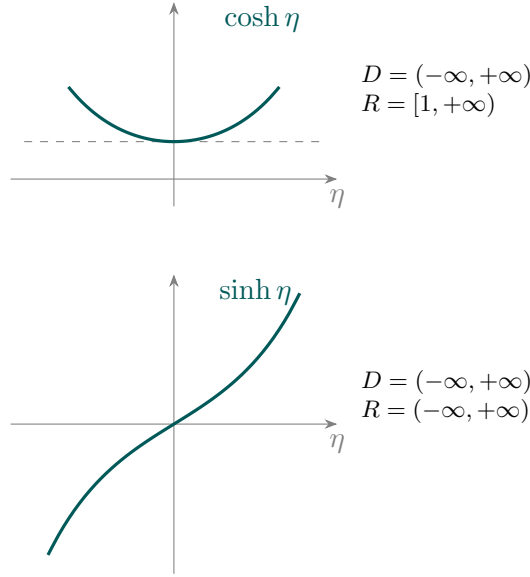
Added explanation (not in the original notes)

The resummation works as for rotations but with one crucial sign change: with $N \equiv -i K_{4\text{vec}}^3$, the nonzero 2×2 block (in the t - z plane) satisfies $N^2 = +\mathbb{K}$ instead of $-\mathbb{K}$, so the even/odd split of the series produces $\cosh$ and $\sinh$ instead of $\cos$ and $\sin$. That single sign traces back to the Lorentzian metric: boosts are “hyperbolic rotations” in a time–space plane. Consistency check: $\cosh^2 \eta - \sinh^2 \eta = 1$ guarantees $\Lambda_B^T g \Lambda_B = g$, and $\Lambda_0^0 = \cosh \eta \geq 1$ with $\det \Lambda_B = +1$ confirms $\Lambda_B \in SO^\uparrow(1, 3)$ for every $\eta \in \mathbb{R}$ — boosts never leave the identity component.

[p. 43] For instance, we get

$$\Lambda_B(\eta, \hat{z}) \begin{pmatrix} m \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} m \cosh \eta \\ 0 \\ 0 \\ m \sinh \eta \end{pmatrix} \equiv \begin{pmatrix} E \\ 0 \\ 0 \\ \pm |\vec{p}| \end{pmatrix}.$$

Remember the graphs of hyperbolic functions:



Depending on η , $\sinh \eta$ can be either negative or positive, corresponding to the possibility of boosting along $\hat{z}$ or $-\hat{z}$.

$$\cosh \eta = \frac{E}{m}, \quad |\sinh \eta| = \frac{|\vec{p}|}{m}.$$

More in general,

$$\Lambda_B(\eta, \hat{p}) \begin{pmatrix} m \\ \vec{0} \end{pmatrix} = \begin{pmatrix} E \\ \vec{p} \end{pmatrix}, \quad \begin{cases} E = m \cosh \eta \\ |\vec{p}| = m |\sinh \eta| \end{cases}$$

Added explanation (not in the original notes)

Rapidity dictionary. Comparing with the familiar form of a boost with velocity $\beta = v/c$: $\cosh \eta = \gamma$, $\sinh \eta = \gamma\beta$, hence $\tanh \eta = \beta$ and $E^2 - |\vec{p}|^2 = m^2(\cosh^2 \eta - \sinh^2 \eta) = m^2$ — the boost moves the four-momentum along the *mass shell*, the hyperbola playing the role that the circle $|\vec{p}| = p$ played for $SE(2)$. The reason rapidity is the natural parameter: successive boosts along the same axis simply *add* their rapidities, $\Lambda_B(\eta_1, \hat{z})\Lambda_B(\eta_2, \hat{z}) = \Lambda_B(\eta_1 + \eta_2, \hat{z})$ (velocities instead combine through the awkward relativistic addition formula). Note that, unlike an angle, η ranges over all of $\mathbb{R}$, not a compact interval — this non-compactness of the Lorentz group is precisely why its interesting unitary representations will be infinite-dimensional.

10 The Lie algebra of the Lorentz group in tensor notation [p. 44]

$$D(\vec{\theta}, \vec{\eta}) = \exp(-i \vec{\theta} \cdot \vec{J} - i \vec{\eta} \cdot \vec{K})$$

$\hookrightarrow$ we “pack” the 6 generators $\vec{J}$ and $\vec{K}$ into an antisymmetric tensor $J^{\mu\nu}$, and the 6 parameters $\vec{\theta}$ and $\vec{\eta}$ into an antisymmetric object $\omega_{\mu\nu}$. We write

$$\begin{aligned} \exp\left(-\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}\right) &= \exp\left(-\frac{i}{2} \omega_{ij} J^{ij} - i \omega_{0i} J^{0i}\right) \\ &= \exp\left(-\frac{i}{2} \omega_{ij} J^{ij} + i \omega_{i0} J^{0i}\right) = \exp\left(-\frac{i}{2} \omega_{ij} J^{ij} - i \omega^{i0} J^{0i}\right) \stackrel{*}{=} \end{aligned}$$

We now define

$$\theta^k \equiv \frac{1}{2} \varepsilon_{kij} \omega^{ij}, \quad J^k \equiv \frac{1}{2} \varepsilon_{kij} J^{ij}, \quad \omega^{i0} = \omega_{0i} \equiv \eta^i, \quad J^{0i} \equiv K^i.$$

Added explanation (not in the original notes)

Why this repackaging works and why it is done. An antisymmetric 4×4 tensor has $\frac{4 \cdot 3}{2} = 6$ independent components — exactly the number of Lorentz generators. Its purely spatial part J^{ij} (3 components) encodes the rotations, with $J^{12} = J^3$ etc. via the ε -symbol (inverse relation: $J^{ij} = \varepsilon_{ijk} J^k$); its time-space part J^{0i} (3 components) encodes the boosts. The factor $\frac{1}{2}$ in $-\frac{i}{2} \omega_{\mu\nu} J^{\mu\nu}$ compensates the double counting of each antisymmetric pair $(\mu\nu)$ and $(\nu\mu)$ in the summed expression. The intermediate manipulations in the displayed chain use only the antisymmetry $\omega_{i0} = -\omega_{0i}$ together with the index raising $\omega^{i0} = g^{ii'} g^{00} \omega_{i'0} = -\omega_{i0}$ (spatial indices flip sign, $\mu = 0$ does not, with this metric). The payoff of the covariant packaging: the six commutators $[J, J]$, $[J, K]$, $[K, K]$ compress into the single manifestly covariant bracket $[J^{\mu\nu}, J^{\rho\sigma}] = i(g^{\nu\rho} J^{\mu\sigma} - g^{\mu\rho} J^{\nu\sigma} - g^{\nu\sigma} J^{\mu\rho} + g^{\mu\sigma} J^{\nu\rho})$, and adding the translation generators P^μ will complete it to the full Poincaré algebra — presumably the next step of the course.

(End of Lecture 2 — the notes continue with the Lorentz algebra in tensor notation and the full Poincaré algebra.)